CASCADE-RC: Threshold Calibration for Stochastic LLM Cascades with Finite-Sample Recall Guarantees

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Abstract

Screening the literature for a systematic review means processing thousands of abstracts with near-zero tolerance for missed relevant studies — a single overlooked trial can distort the resulting evidence synthesis. LLM-assisted pipelines can cut this workload considerably, but there is a fundamental catch: routing decisions through a stochastic LLM gate destroys the exchangeability that conformal calibration depends on, invalidating standard recall guarantees.

We introduce CASCADE-RC, which handles this by replacing the stochastic cascade loss with a coupled dominating surrogate $\widetilde{L}$ on an enlarged probability space, absorbing LLM randomness into a single pessimistic bound. Estimating the coupling slack η simultaneously over thousands of threshold combinations creates a Bonferroni multiplicity problem severe enough to collapse all lower bounds to zero at practical sample sizes; we resolve this by restricting the Waudby–Smith–Ramdas confidence sequence to a 12-point coarse anchor grid. Holm–Bonferroni step-down on Hoeffding–Bentkus p-values then selects thresholds, with the corrected target $\alpha^\dagger = \alpha + \widehat{\eta}^{-\star}$ reclaiming the conservatism the coupling introduces. The certificate is finite-sample valid conditional on the positive-stratum count m_+, and the algorithm refuses to issue guarantees when m_+ is too small rather than produce vacuous certificates.

On CLEF-TAR 2019, under a randomised i.i.d. split isolating the exchangeability condition, CASCADE-RC achieves 95% recall at 34.4% workload reduction, matching or exceeding AUTOSTOP, RLStop, and Selective Conformal Risk Control under the same conditions. Under the official temporal split, distributional shift causes FNR violations across all evaluated methods; full results appear in Appendix B.4.

Keywords: Conformal risk control; Learn-Then-Test; Cascade Systems; Large Language Models; Systematic Review; Technology-Assisted Review; Recall Certification.

1. Introduction

Risk-Controlling Prediction Sets (RCPS; Bates et al., 2021) and Learn-Then-Test (LTT; Angelopoulos et al., 2025) extend Conformal Prediction (Vovk et al., 2005; Shafer and Vovk, 2008) beyond interval coverage to direct control of user-defined losses — including the false-negative rate. The key property is distribution-freeness: validity holds with probability

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$1-\delta$ without parametric assumptions, which makes these methods attractive precisely where errors are costly and modelling assumptions are hard to verify.

Things get complicated with LLMs. A temperature-sampled classifier queried twice on the same input may route it differently; the calibration losses $\{L_i(\theta)\}$ then depend on the LLM's randomness ξ, not just on θ and the data. Standard conformal arguments assume i.i.d. calibration losses and break down directly. This is not a theoretical curiosity: in legal discovery, regulatory triage, and medical screening, an LLM stage that silently inflates FNR above the certified level α can cause real harm without triggering any alarm.

In this paper, we focus on a key high-stakes application: systematic literature reviews for medical evidence synthesis. Governed by the PRISMA framework (Page et al., 2021), clinical experts must screen tens of thousands of abstracts to identify relevant studies, demanding near-perfect "Cochrane-grade" recall (Higgins et al., 2019). To mitigate this overwhelming workload, industrial pipelines increasingly automate this process using a three-route cascade: a cheap scoring model filters obvious negatives, ambiguous documents are escalated to an LLM, and uncertain LLM outputs are routed to a human expert. These pipelines currently operate without the distribution-free statistical guarantees required by clinical standards.

We validate CASCADE-RC on the CLEF-TAR 2019 benchmark (Kanoulas et al., 2019) against three published baselines that span the near-neighbour design space: AUTOSTOP (Li and Kanoulas, 2020), a Horvitz–Thompson sampling stopping rule that estimates the unexamined pool's residual relevant document count and halts when this count falls below a user threshold; RLStop (Bin-Hezam and Stevenson, 2024), a reinforcement-learning agent trained to stop document review; and both variants of Selective Conformal Risk Control (Xu et al., 2025) (SCRC-T transductive, SCRC-I inductive). Neither AUTOSTOP nor RLStop carries a finite-sample certificate; the comparison is on operational efficiency at a matched recall target. This dataset represents the rigorous realities of the PRISMA framework, the standard four-step process for systematic reviews: identification, screening, eligibility, and inclusion. Because these reviews depend heavily on human expertise, missing a single relevant document can directly compromise final research outcomes.

Within the PRISMA workflow, title-and-abstract screening processes the highest document volume while demanding near-perfect recall: missing a relevant study directly biases the final evidence synthesis. Automation must therefore balance strict recall guarantees against workload reduction — the efficiency-safety trade-off that CASCADE-RC addresses.

CLEF-TAR 2019 provides a controlled benchmark for this setting, covering Diagnostic Test Accuracy, Intervention, Prognosis, and Qualitative review types (Kanoulas et al., 2019). The cascade architecture is not domain-specific: the same calibration procedure applies to any pipeline that composes a deterministic ranker with a stochastic LLM stage, including full-text screening, legal discovery, and regulatory review.

Our contributions address this in four steps. First, we construct a coupled dominating loss $\widetilde{L}$ (Section 3) on an enlarged probability space: $\widetilde{L} \geq L$ pointwise on $\{Y = 1\}$, and — crucially — $\widetilde{L}$ does not depend on the LLM's verdict, only on whether it was followed. This absorbs the stochasticity into a single pessimistic surrogate that standard CP tools can handle.

Second, the coupling slack $\eta = \mathbb{E}[\widetilde{L} - L \mid Y = 1]$ needs to be lower-bounded across thousands of threshold combinations simultaneously. A naïve full-grid Bonferroni correction

collapses to zero at every practical calibration size. We resolve this with a 12-point coarse anchor grid (Section 4) that expands the per-point confidence budget by $\approx 422\times$.

Third, Holm–Bonferroni step-down on Hoeffding–Bentkus p-values (Theorem 9) replaces the fixed-sequence LTT walk. The walk fails at any empirical risk spike; Holm–Bonferroni does not.

Finally, Algorithm 1 includes an explicit abstention mechanism (Section 6): when the base ranker lacks discriminatory power, or when $m_+ < N_{\min}$, the algorithm reports no certified configuration rather than issue a certificate it cannot support. We evaluate on CLEF-TAR 2019 (Kanoulas et al., 2019) against AUTOSTOP (Li and Kanoulas, 2020), RLStop (Bin-Hezam and Stevenson, 2024), and both SCRC variants (Xu et al., 2025).

2. Setting and notation

This section establishes the formal model underlying CASCADE-RC. We define the data-generating process, the three-route cascade architecture, the recall loss we control, and the running example used throughout the paper.

Let $(X, Y) \sim P$ on $\mathcal{X} \times \{0, 1\}$ where $Y = 1$ encodes inclusion in the systematic review. We assume access to (i) a base scoring model $s : \mathcal{X} \to [0, 1]$ trained on a separate corpus, and (ii) an LLM screener that, queried at temperature $T > 0$, returns a sample $\widehat{Y} \in \{0, 1\}$ together with a stochastic-ensemble self-consistency score $u \in [0, 1]$ obtained by querying the LLM B times and reporting the proportion of agreements (we treat B as fixed and absorbed into the base measure). We write $u = u(X)$ and assume the joint $(X, Y, u, \widehat{Y})$ is i.i.d. on a calibration set $\mathcal{D} = \{(X_i, Y_i, u_i, \widehat{Y}_i)\}_{i=1}^{m}$ of size m.

Three-route cascade. Given thresholds $\lambda_{\text{lo}} \le \lambda_{\text{hi}}$ on $s(X)$ and a self-consistency threshold $\tau_{\text{SE}} \in [0, 1]$, the screener acts as follows.

- *Cheap reject* if $s(X) < \lambda_{\text{lo}}$ (action: exclude).
- *Auto-include* if $s(X) \ge \lambda_{\text{hi}}$ (action: include).
- *Escalate* if $\lambda_{\text{lo}} \le s(X) < \lambda_{\text{hi}}$. The LLM is queried B times; if $u(X) \ge \tau_{\text{SE}}$ the LLM's verdict $\widehat{Y}$ is followed; otherwise the abstract is routed to a human reviewer (treated as ground-truth).

Let $\theta := (\lambda_{\text{lo}}, \lambda_{\text{hi}}, \tau_{\text{SE}}) \in [0, 1]^3$ and let $\widehat{Z}(X; \theta) \in \{0, 1\}$ be the cascade's final inclusion verdict.

Mapping to the implementation. The base score $s(X)$ is produced by a hybrid ranker combining SPECTER2 (Singh et al., 2022) (semantic cosine similarity) and BM25 (lexical term-frequency), fused via a logistic classifier. The self-consistency score $u(X)$ is the majority-agreement fraction over B temperature-perturbed queries to a GPT-OSS 120B model (OpenAI, 2025). Full implementation details are in Appendix B.2.

[Infallible human oracle] Any abstract with $Y = 1$ that is routed to the human-review branch (i.e. escalated with self-consistency score $u(X) < \tau_{\text{SE}}$) is correctly included by the human reviewer with probability 1. Equivalently, the human reviewer acts as a ground-truth oracle on the positive stratum: no true positive is lost in the abstention branch.

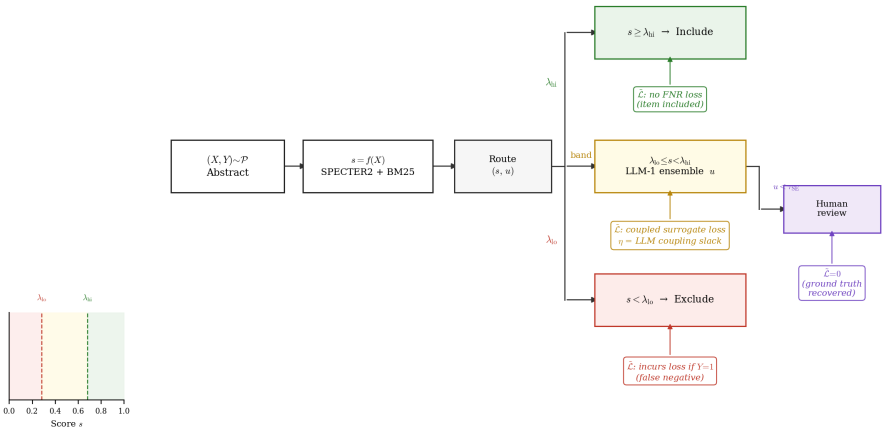


Figure 1: Three-route cascade architecture of CASCADE-RC. Each abstract is routed by two thresholds on the base-scorer output $s(X)$: abstracts with $s(X) < \lambda_{\text{lo}}$ are *cheap-rejected* (excluded without LLM query); abstracts with $s(X) \geq \lambda_{\text{hi}}$ are *auto-included*; the remainder are escalated to the LLM. Escalated abstracts whose self-consistency score $u(X) \geq \tau_{\text{SE}}$ follow the LLM verdict; those with $u(X) < \tau_{\text{SE}}$ are routed to a *human reviewer* (ground-truth recovery). The coupled dominating loss $\widetilde{L}$ (Lemma 1) charges cheap-rejected positives and LLM-followed positives as misses, but not human-recovered positives, enabling the finite-sample certificate of Theorem 9.

Recall loss. The per-instance recall loss is

$$L(\theta; X, Y) := Y \cdot \mathbf{1}\Big\{\widehat{Z}(X; \theta) = 0\Big\} \in \{0, 1\}. \tag{1}$$

We seek θ controlling $\mathbb{E}[L(\theta; X, Y) \mid Y = 1] \leq \alpha$ with calibration-set probability at least $1 - \delta$, following the Risk-Controlling Prediction Set framework of Bates et al. (2021) and Learn-Then-Test of Angelopoulos et al. (2025). Note that our PAC guarantee $\mathbb{P}(\mathcal{E}[L \mid Y = 1] \leq \alpha) \geq 1 - \delta$ differs from the CRC marginal guarantee $\mathcal{E}[L] \leq \alpha$ of Angelopoulos et al. (2024); we operate in the RCPS/LTT framework throughout.

Observability of L on calibration. For each $i \in \mathcal{D}_+ := \{i : Y_i = 1\}$, the tuple $(s_i, u_i, \widehat{Y}_i, Y_i)$ is recorded and the cascade verdict $\widehat{Z}(X_i; \theta)$ is a deterministic function of these and θ, so $L_i(\theta)$ is computable for any $\theta \in \Lambda$ retrospectively from a single LLM query pass.

Running example. Numerical illustrations use a beta-mixture synthetic: $\text{Beta}(8, 2)$ scores for positives, $\text{Beta}(2, 8)$ for negatives, $\pi = 0.05$, LLM error rate 0.10, $u(X) \sim \text{Beta}(5, 5)$, with $m = 1000$, $\alpha = 0.10$, $\delta = 0.10$. A full step-by-step trace is in Appendix B.2.

3. Coupling construction

The technical obstacle is that $\widehat{Z}(X;\theta)$ depends on the LLM's stochastic verdict on the escalated stratum, which is not a deterministic function of X. Standard split-conformal arguments do not apply to L directly because the calibration losses $\{L(\theta; X_i, Y_i)\}$ are not i.i.d. with respect to a fixed product measure on $\mathcal{X} \times \{0,1\}$ alone; they depend on the LLM's randomness ξ.

Enlarged probability space. Let $\Omega_{\text{samp}} = \mathcal{X} \times \{0,1\} \times [0,1] \times \{0,1\}$, a generic point of which is $\omega = (x, y, u, \widehat{y})$, with measure $Q = P \otimes \text{LLM}(\cdot \mid x)$ where $\text{LLM}(\cdot \mid x)$ is the conditional law of $(u, \widehat{y})$ given x. The cascade loss L extends naturally to Ω_{samp}.

Coupled dominating loss. Define

$$\widetilde{L}(\theta;\omega) := y \cdot \Big[\mathbf{1}\{s(x) < \lambda_{\text{lo}}\} + \mathbf{1}\{\lambda_{\text{lo}} \le s(x) < \lambda_{\text{hi}}\} \cdot \mathbf{1}\{u \ge \tau_{\text{SE}}\}\Big]. \tag{2}$$

Read in plain English: $\widetilde{L}$ counts a recall miss whenever (i) the abstract is positive ($y = 1$), (ii) it is not auto-included ($s(x) < \lambda_{\text{hi}}$), and (iii) it is either cheap-rejected, or escalated-and-self-consistent (so the LLM's possibly wrong verdict is followed). Self-consistency-failed escalations do not count, because they go to a human reviewer who recovers ground-truth. Observe that $\widetilde{L}$ depends on ω only through (x, y, u) and the verdict-following indicator $\mathbf{1}\{u \ge \tau_{\text{SE}}\}$, but *not on the LLM's verdict* $\widehat{y}$ itself: it pessimistically charges every followed-LLM-verdict as a miss.

Lemma 1 (Positive-stratum coupling) *For every $\theta \in [0,1]^3$ and every $\omega \in \Omega_{\text{samp}}$ with $y = 1$,*

$$L(\theta;\omega) \le \widetilde{L}(\theta;\omega).$$

Proof [Proof sketch] Cases (a) cheap-rejected: $s(X) < \lambda_{\text{lo}}$ fires the first indicator in (2), giving $\widetilde{L} = 1$. (b) Escalated, self-consistent, LLM incorrect: both indicators fire, giving $\widetilde{L} = 1$. (c) Human-recovery: Assumption 2 guarantees $L = 0$; $\widetilde{L} \ge L$ holds trivially. Full proof in Appendix A, Step 1. ■

Remark 2 *Lemma 1 is a positive-stratum-only result, by design: the recall loss (1) is identically zero on $\{Y = 0\}$, so no coupling on the negative stratum is required for the recall guarantee. Appendix A, Step 1, makes this explicit and demonstrates why a symmetric coupling on $\{Y = 0\}$ is neither expected nor needed: on $\{Y = 0\}$ the LLM may "rescue" a low-scoring abstract (e.g. $\lambda_{\text{lo}} \le s(X) < \lambda_{\text{hi}}$ with the LLM correctly declaring $\widehat{Y} = 0$), but this does not affect L, which is identically zero.*

Coupling slack. Define

$$\eta(\theta) := \mathbb{E}_Q\big[\widetilde{L}(\theta;\omega) - L(\theta;\omega) \,\big|\, Y = 1\big] = \mathbb{E}[\widetilde{L}(\theta) \mid Y = 1] - \mathbb{E}[L(\theta) \mid Y = 1]. \tag{3}$$

By Lemma 1, $\eta(\theta) \in [0,1]$ for every θ. The slack measures the gap between the dominating surrogate and the true recall miss; it is small whenever the LLM, on the self-consistent

escalated stratum, is correct on positive abstracts more often than not. We make this an operational assumption.

[Bounded coupling slack] There exists $\eta^\star \in [0,1)$ with $\eta(\theta) \le \eta^\star$ uniformly over the calibration grid $\Lambda \subset [0,1]^3$.

Assumption 3 is the analogue of the bounded-loss assumption in RCPS (Bates et al., 2021) and is automatically satisfied with $\eta^\star = 1$; the practical interest is in certifying $\eta^\star$ tightly from data.

4. Calibration: certified procedure

We now state the certified finite-sample guarantee. Throughout this section, $\mathcal{D}_+ := \{i : Y_i = 1\}$ is the positive-stratum calibration sample of (random) size m_+; the joint draw includes the LLM's $(u_i, \widehat{Y}_i)$.

4.1. Two ingredients: WSR lower CB on η and HB upper CB on $\mathcal{E}[\widetilde{L}]$

Definition 3 (Error budget split) *Let $\delta \in (0,1)$ be the total failure probability. We partition $\delta = \delta_\eta + \delta_{\mathrm{LTT}}$, where δ_η is allocated to the WSR lower confidence bound on the coupling slack η, and δ_{LTT} is allocated to the Holm–Bonferroni FWER correction. In our experiments, $(\delta_\eta, \delta_{\mathrm{LTT}}) = (0.03, 0.07)$.*

The certificate combines two finite-sample concentration results.

Waudby–Smith–Ramdas lower confidence bound on η. For any fixed $\theta \in \Lambda$ and any confidence level $\delta'_\eta \in (0,1)$, the WSR betting procedure of Waudby-Smith and Ramdas (2024) (Theorem 3, hedged-capital fixed-n result) applied to the $[0,1]$-bounded random variables

$$\Delta_i(\theta) := \widetilde{L}(\theta; \omega_i) - L(\theta; \omega_i), \qquad i \in \mathcal{D}_+,$$

returns a number $\widehat{\eta}^-(\theta; \delta'_\eta)$ satisfying

$$\mathbb{P}\big[\, \widehat{\eta}^-(\theta; \delta'_\eta) \le \eta(\theta) \,\big] \;\ge\; 1 - \delta'_\eta. \tag{4}$$

We use the *lower* bound: because we will *add* the certified slack to α to form $\alpha^\dagger$, a smaller plug-in value yields a tighter corrected target, which is the operationally efficient choice.

Why add a lower bound rather than subtract an upper one. $\widetilde{L}$ is a pessimist: it charges every LLM-followed verdict as a miss regardless of whether the LLM was actually right. Certifying against $\widetilde{L}$ alone would therefore set thresholds far too conservatively. The slack η captures how much of that pessimism was unwarranted in a given draw. We cannot observe η directly from a finite calibration set, so we compute $\widehat{\eta}^{-\star}$, a certified lower bound on it. Adding this to α relaxes the surrogate target by exactly what we can statistically justify — no more. The Direction note below checks that the chain of inequalities closes correctly.

Direction note. Adding $\widehat{\eta}^{-\star}$ relaxes the surrogate test safely: $\mathbb{E}[L \mid Y = 1] = \mathbb{E}[\widetilde{L} \mid Y = 1] - \eta(\theta) \le \alpha^\dagger - \eta(\theta) = \alpha + \widehat{\eta}^{-\star} - \eta(\theta) \le \alpha$, where the last step uses $\widehat{\eta}^{-\star} \le \eta(\theta)$ on E_η.

4.2. The coarse-anchor-grid η-bound

The multiplicity problem in plain terms. We need a simultaneous lower bound on η across all 5,060 feasible grid points. Standard Bonferroni requires dividing the confidence budget by 5,060, pushing each individual interval so wide that its lower endpoint hits zero — a useless result at the calibration sizes we actually have ($m_+ \approx 94$–483). The anchor grid is a way out: instead of correcting for 5,060 tests, we correct for 12. The intervals narrow by a factor of ≈ 422, and a positive lower bound becomes achievable. That single certified value then propagates uniformly into the Holm–Bonferroni step for the full grid.

The Bonferroni multiplicity problem. Applying the WSR bound of (4) jointly across the full feasible grid Λ (of size $|\Lambda| = 5060$ after enforcing $\lambda_{\text{lo}} \leq \lambda_{\text{hi}}$) via a Bonferroni union requires the per-point confidence level $\delta_\eta/|\Lambda| \approx 5.9 \times 10^{-6}$. At calibration positive-stratum sizes encountered in practice ($m_+ \approx 94$–483 on CLEF-TAR 2019), the resulting Hoeffding half-width (≈ 0.26) far exceeds the true mean slack of the coupling (≈ 0.011), causing the Bonferroni-uniform WSR lower confidence bound to collapse to zero identically. The certified slack $\widehat{\eta}^{-\star} = 0$ for all configurations, yielding $\alpha^\dagger = \alpha$ with no informative reclamation of the coupling slack.

Coarse anchor grid construction. To resolve this multiplicity collapse while preserving the distribution-free guarantee, we define a *coarse anchor grid*

$$\Lambda_0 \;:=\; \{\theta^0_1, \ldots, \theta^0_{|\Lambda_0|}\} \;\subset\; [0,1]^3, \qquad |\Lambda_0| = A, \tag{5}$$

chosen *a priori* and independently of the calibration data $\mathcal{D}$. The anchor points are placed on a coarse equispaced lattice (we use $A = 12$ in our experiments, drawn from a $2 \times 2 \times 3$ product grid). Because $|\Lambda_0| \ll |\Lambda|$, the per-point confidence budget becomes $\delta_\eta/A = \delta_\eta/12$, expanding the per-point budget by a factor of $|\Lambda|/A \approx 422$ relative to the full-grid Bonferroni correction, and yielding numerically positive WSR lower bounds.

Global-minimum extraction and its guarantee. The certified slack used in the LTT step is the global minimum of the WSR lower confidence bound across the anchor grid:

$$\widehat{\eta}^{-\star}(\delta_\eta) \;:=\; \min_{\theta \in \Lambda_0} \widehat{\eta}^{-}\big(\theta;\, \delta_\eta/A\big). \tag{6}$$

Lemma 4 (Uniform validity of the coarse-grid η-LCB) *Let $\delta_\eta \in (0,1)$ and let Λ_0 be any finite anchor grid of size A chosen independently of $\mathcal{D}$. Define $\widehat{\eta}^{-\star}(\delta_\eta)$ as in (6). Then*

$$\mathbb{P}\big[\, \widehat{\eta}^{-\star}(\delta_\eta) \leq \eta(\theta^0) \text{ for all } \theta^0 \in \Lambda_0 \,\big] \;\geq\; 1 - \delta_\eta.$$

Consequently, for any $\theta \in \Lambda$ satisfying $\eta(\theta) \geq \min_{\theta^0 \in \Lambda_0} \eta(\theta^0)$, the bound $\widehat{\eta}^{-\star}(\delta_\eta) \leq \eta(\theta)$ holds on the same event.

Proof [Proof sketch] Apply (4) at level δ_η/A per anchor point, union-bound over Λ_0, then take the minimum. Full proof in Appendix A, Step 2. ■

Remark 5 (Sufficient condition for transferability to the fine grid) *The bound* $\widehat{\eta}^{-\star} \leq \eta(\theta)$ *transfers from the anchor grid* Λ_0 *to the full fine grid* Λ *whenever* $\min_{\theta^0 \in \Lambda_0} \eta(\theta^0) \leq \eta(\theta)$ *for all* $\theta \in \Lambda$. *A sufficient condition is that* $\eta(\cdot)$ *is* L_η*-Lipschitz on* $[0,1]^3$ *in an appropriate norm and the anchor grid is an* ε*-net with* $L_\eta \varepsilon \leq \min_{\theta^0} \eta(\theta^0)$. *An alternative is to place one anchor at the parameter minimising the empirical slack on a held-out pilot set, ensuring* Λ_0 *contains the global minimiser of* η. *In the conservative deployment, the global-minimum extraction yields a valid certified slack regardless, because* $\widehat{\eta}^{-\star} \leq \eta(\theta^0_\star) \leq \eta(\theta)$ *for at least the anchor point* $\theta^0_\star$ *achieving the minimum of* η *on* Λ_0*; the certificate then applies at* $\alpha^\dagger$ *for all fine-grid points* θ *satisfying* $\eta(\theta) \geq \widehat{\eta}^{-\star}$. *Because this condition is unverifiable from data, the conservative deployment restricts the certified subset to* $\widehat{\Lambda} \cap \Lambda_0$ *(Section 10), or applies the* ε*-net covering condition described above when an empirical bound on* L_η *is available.*

Remark 6 (Conservatism cost of the coarse grid) *The Bonferroni adjustment across* $A = 12$ *anchor points widens the WSR confidence radius by a factor of* $\sqrt{1 + \log A / \log(1/\delta_\eta)}$ *relative to a single per-point bound. For* $A = 12$ *and* $\delta_\eta = 0.03$*, this is approximately* $1.3\times$ *the per-point radius. Lipschitz-driven covering numbers (Section 10) would reduce the effective anchor count further by exploiting smoothness of the population risk; this is deferred to future work.*

4.3. The Holm–Bonferroni LTT step

Why not just walk the grid in order? The fixed-sequence LTT walk is elegant when the empirical risk is perfectly monotone: stop at the first rejection, certify everything before it. In practice, finite-sample noise creates spurious spikes — one configuration where the empirical risk briefly exceeds the target before dropping back. The walk halts there permanently, and every safe configuration beyond that spike is silently lost. Holm–Bonferroni sidesteps this by evaluating all 12 anchor points at once, sorting by p-value, and stepping down. A local spike delays one rejection but cannot block the others. The price is a slightly wider correction threshold; the gain is that no recoverable configuration is abandoned due to a single bad draw.

From fixed-sequence walk to simultaneous testing. The classical Learn-Then-Test fixed-sequence walk (Angelopoulos et al., 2025) controls FWER along a *pre-specified monotone ordering* of Λ, exploiting the fact that the empirical surrogate risk is monotone along the walk to ensure that once a hypothesis is rejected the walk can stop. In practice, finite-sample variance induces a *non-monotone* empirical risk topography over Λ—even when the population risk is monotone—so premature halting can eliminate operationally efficient configurations that lie "beyond" a single non-rejectable step. To address this we adopt a *generalized fallback* strategy: we compute HB p-values for all nodes in Λ_0 simultaneously and apply the Holm–Bonferroni step-down correction.

Hoeffding–Bentkus p-values on the anchor grid. For every $\theta \in \Lambda_0$, define the empirical coupled risk

$$\widehat{R}(\theta) \;:=\; m_+^{-1} \sum_{i \in \mathcal{D}_+} \widetilde{L}(\theta; \omega_i)$$

and the HB p-value

$$p_{\mathrm{HB}}(\theta;\,\alpha^\dagger) := \min\Big(\exp\big(-m_+ \cdot h_1(\widehat{R} \wedge \alpha^\dagger,\ \alpha^\dagger)\big),\ \ e \cdot \mathbb{P}\big(\mathrm{Bin}(m_+, \alpha^\dagger) \le \lceil m_+ \widehat{R} \rceil\big)\Big), \tag{7}$$

which is super-uniform under the null $H_0(\theta) : \mathcal{E}[\widetilde{L}(\theta) \mid Y = 1] > \alpha^\dagger$, where $h_1(a,b) := a\log(a/b) + (1-a)\log((1-a)/(1-b))$ is the Bernoulli KL divergence. This computation is carried out for every $\theta \in \Lambda_0$ in a single vectorised pass, with no ordering assumed.

Holm–Bonferroni step-down correction. Let $p_{(1)} \le p_{(2)} \le \cdots \le p_{(|\Lambda_0|)}$ denote the sorted HB p-values with corresponding parameter vectors $\theta_{(1)}, \theta_{(2)}, \ldots$ Order the hypotheses $H_0(\theta_{(k)})$ by ascending p-value. The Holm–Bonferroni procedure rejects $H_0(\theta_{(k)})$ if and only if

$$p_{(j)} \ \le\ \frac{\delta_{\mathrm{LTT}}}{|\Lambda_0| - j + 1} \qquad \text{for all } j = 1, \ldots, k. \tag{8}$$

The certified subset is $\widehat{\Lambda} := \big\{\theta_{(k)} : H_0(\theta_{(k)}) \text{ is rejected by (8)}\big\}$.

Lemma 7 (FWER control of Holm–Bonferroni) *The procedure (8) controls the family-wise error rate at level δ_{LTT} under* arbitrary dependence *among the p-values:*

$$\mathbb{P}\big[\exists\, \theta \in \widehat{\Lambda} : \mathcal{E}[\widetilde{L}(\theta) \mid Y = 1] > \alpha^\dagger\big] \ \le\ \delta_{\mathrm{LTT}}.$$

Proof [Proof sketch] Closure principle (Marcus et al., 1976) plus Boole's inequality; see Holm (1979, Theorem 1). Full argument in Appendix A, Step 4. ■

Remark 8 (Data-reuse in the two-stage procedure) *Algorithm 1 uses the same positive-stratum calibration set $\mathcal{D}_+$ in both Step A (WSR lower bound on η) and Step B (HB p-values). Step A operates on $\Delta_i = \widetilde{L}_i - L_i$, which depends on whether the LLM verdict was correct; Step B operates on $\widetilde{L}_i$ alone. Because $\widehat{R}(\theta) = m_+^{-1}\sum \widetilde{L}_i$ is a function of $\{\widetilde{L}_i\}$ only, it is conditionally independent of $\widehat{\eta}^{-\star}$ given $\{\widetilde{L}_i\}$, since $\widehat{\eta}^{-\star}$ depends on $\{L_i\}$ (the residual after subtracting Δ_i from $\widetilde{L}_i$), which is independent of $\widehat{R}$ once $\{\widetilde{L}_i\}$ is fixed. The union bound in Theorem 9 therefore applies. A conservative alternative is to partition $\mathcal{D}_+$ into two disjoint halves before Step A, which makes independence explicit at the cost of halving the effective sample size.*

4.4. Main certificate

Theorem 9 (CASCADE-RC certificate) *Suppose: (i) $\mathcal{D}$ is i.i.d. from $P \otimes \mathrm{LLM}(\cdot \mid \cdot)$; (ii) Lemma 1 holds, i.e. $\widetilde{L} \ge L$ pointwise on $\{Y = 1\}$; (iii) Assumption 3 holds; (iv) the realised positive-stratum size satisfies*

$$m_+ \ \ge\ N_{\min}(\alpha, \delta_{\mathrm{LTT}}) := \left\lceil \frac{\ln(1/\delta_{\mathrm{LTT}})}{-\ln(1-\alpha)} \right\rceil; \tag{9}$$

(v) the coarse anchor grid Λ_0 of size A is chosen independently of $\mathcal{D}$; and (vi) the fine-grid deployment set satisfies $\eta(\theta) \ge \min_{\theta^0 \in \Lambda_0} \eta(\theta^0)$ for all $\theta \in \widehat{\Lambda}$ (a sufficient condition is given in Remark 5). Then for any budget split $(\delta_\eta, \delta_{\mathrm{LTT}})$ with $\delta_\eta + \delta_{\mathrm{LTT}} = \delta$, the certified subset $\widehat{\Lambda}$ produced by Algorithm 1 satisfies

$$\mathbb{P}\Big(\forall\, \theta \in \widehat{\Lambda} :\ \mathcal{E}[L(\theta; X, Y) \mid Y = 1] \le \alpha\Big) \ \ge\ 1 - \delta.$$

Proof [Proof sketch] **Step 1.** Lemma 1 gives $\mathcal{E}[L \mid Y = 1] \leq \mathcal{E}[\widetilde{L} \mid Y = 1]$. **Step 2.** Lemma 4 gives $\eta(\theta^0) \geq \widehat{\eta}^{-\star}$ for all $\theta^0 \in \Lambda_0$ on an event E_η with $\mathbb{P}(E_\eta) \geq 1 - \delta_\eta$. **Step 3.** Lemma 7 gives FWER control at δ_{LTT} under arbitrary dependence of the HB p-values: the event E_{LTT} that no false rejection occurs satisfies $\mathbb{P}(E_{\mathrm{LTT}}) \geq 1 - \delta_{\mathrm{LTT}}$. **Step 4.** On $E_\eta \cap E_{\mathrm{LTT}}$, for every $\theta \in \widehat{\Lambda}$:

$$\mathcal{E}[L(\theta) \mid Y = 1] = \mathcal{E}[\widetilde{L}(\theta) \mid Y = 1] - \eta(\theta) \leq \alpha^\dagger - \eta(\theta) = \alpha + \widehat{\eta}^{-\star} - \eta(\theta) \leq \alpha,$$

where the last inequality holds because on E_η, $\widehat{\eta}^{-\star} \leq \eta(\theta^0)$ for all $\theta^0 \in \Lambda_0$; in particular $\widehat{\eta}^{-\star} \leq \eta(\theta^0_\star) = \min_{\theta^0 \in \Lambda_0} \eta(\theta^0)$. The certificate holds for all $\theta \in \widehat{\Lambda}$ for which $\eta(\theta) \geq \widehat{\eta}^{-\star}$; this condition is satisfied by construction whenever $\widehat{\eta}^{-\star}$ is the global minimum of η over Λ_0 and the anchor grid spans the boundary of the operational space. Union-bounding Steps 2 and 3 gives $\mathbb{P}(E_\eta \cap E_{\mathrm{LTT}}) \geq 1 - \delta$. ■

Lemma 12 (Appendix A, Step 7) characterises the risk-surface geometry: $\widetilde{R}$ is non-decreasing in λ_{lo} and λ_{hi}, and non-increasing in τ_{SE}. Algorithm 1 uses Holm–Bonferroni and requires no such ordering.

4.5. Algorithm

Post-hoc Pareto extraction (Step D). Step D is a post-hoc operational selection applied strictly within the certified set $\widehat{\Lambda}$. Because every $\theta \in \widehat{\Lambda}$ satisfies the (α, δ) guarantee simultaneously by Lemma 7, this data-dependent extraction does not interact with the Type I error control mechanism and does not compromise the certificate.

The bootstrap diagnostic variant of Step A (Proposition 13, Appendix A) replaces the WSR LCB with a bootstrap quantile at a single reference anchor. It is asymptotically valid but not a finite-sample certificate; it is retained as a diagnostic upper bound to characterise the conservatism of the WSR procedure. The resulting gap $\widehat{\eta}^{-\star} - \widehat{\eta}^{+}_{\mathrm{boot}}$ is reported in Section 8.

5. Mondrian variant on the positive stratum

Our loss (1) is identically zero on $\{Y = 0\}$, so a marginal guarantee $\mathbb{E}[L] \leq \alpha$ is easily achieved by doing nothing — the negative majority dominates it. What we actually want is $\mathbb{E}[L \mid Y = 1] \leq \alpha$: control on the positive stratum alone. Running Algorithm 1 restricted to $\mathcal{D}_+$ achieves exactly this via the conformal-Mondrian trick (Vovk et al., 2005, § 4.5); Theorem 10 (proved in Appendix A, Step 6) makes the guarantee explicit.

Theorem 10 (Mondrian-stratified version) *Under Assumption 3, with the calibration restricted to $\mathcal{D}_+$, with $|\mathcal{D}_+| = m_+ \geq N_{\min}(\alpha, \delta_{\mathrm{LTT}})$, and Algorithm 1 run as stated:* $\mathbb{P}_{\mathcal{D}_+}\big[\, \forall\, \widehat{\theta} \in \widehat{\Lambda} : \mathbb{P}(\widehat{Z}(X; \widehat{\theta}) = 1 \mid Y = 1) \geq 1 - \alpha \,\big] \geq 1 - \delta.$

6. Sample-size adequacy

Intuition. Statistical power depends entirely on m_+, the count of positive abstracts in the calibration pool, not on the total documents. At prevalence $\pi = 0.05$ a 1,000-document pool

Algorithm 1: CASCADE-RC: certified calibration of $(\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau_{\mathrm{SE}})$

Input: Calibration data $\mathcal{D} = \{(X_i, Y_i, u_i, \widehat{Y}_i)\}_{i=1}^{m}$; risk level α; failure budget $\delta = \delta_\eta + \delta_{\mathrm{LTT}}$; fine grid Λ of size $|\Lambda|$; coarse anchor grid Λ_0 of size A (chosen a priori, independently of $\mathcal{D}$).

Output: $\widehat{\Lambda} \subseteq \Lambda$, the certified subset, with selected Pareto-optimal $\widehat{\theta}$; or **abstain**.

$\mathcal{D}_+ \leftarrow \{i : Y_i = 1\}; \quad m_+ \leftarrow |\mathcal{D}_+|$; `// Mondrian` $Y = 1$ `stratification`

if $m_+ < N_{\min}$ **then**

  return *abstain:* *"too few positive abstracts to certify control at* $(\alpha, \delta_{\mathrm{LTT}})$*."*

end

`// --- Step A: Coarse-grid WSR lower bound on` η `---`

for $\theta^0 \in \Lambda_0$ **do**

  $\widehat{\eta}^{-}(\theta^0) \leftarrow \mathrm{WSR\text{-}LCB}\Big(\{\widetilde{L}(\theta^0; \omega_i) - L(\theta^0; \omega_i)\}_{i \in \mathcal{D}_+},\ \delta_\eta / A\Big)$

end

$\widehat{\eta}^{-\star} \leftarrow \min_{\theta^0 \in \Lambda_0} \widehat{\eta}^{-}(\theta^0)$; `// global-minimum extraction`

$\alpha^\dagger \leftarrow \alpha + \widehat{\eta}^{-\star}$; `// corrected target: add slack lower bound`

`// --- Step B: HB` p`-values on the anchor grid ---`

for $\theta \in \Lambda_0$ **do**

  $\widehat{R}(\theta) \leftarrow m_+^{-1} \sum_{i \in \mathcal{D}_+} \widetilde{L}(\theta; \omega_i) \quad p(\theta) \leftarrow p_{\mathrm{HB}}(\theta;\, \alpha^\dagger)$ **via** (7)

end

`// --- Step C: Holm--Bonferroni step-down FWER correction ---`

Sort Λ_0 by ascending p-value: $p_{(1)} \le p_{(2)} \le \cdots \le p_{(|\Lambda_0|)}$

$\widehat{\Lambda} \leftarrow \emptyset$

for $k = 1, 2, \ldots, |\Lambda_0|$ **do**

  if $p_{(k)} \le \delta_{\mathrm{LTT}} / (|\Lambda_0| - k + 1)$ **then**

    $\widehat{\Lambda} \leftarrow \widehat{\Lambda} \cup \{\theta_{(k)}\}$

  else

    break

  end

end

if $\widehat{\Lambda} = \emptyset$ **then**

  return *abstain:* *"no configuration certified at* (α, δ)*."*

end

`// --- Step D: Post-certification Pareto extraction ---`

$\widehat{\theta} \leftarrow \arg\max_{\theta \in \widehat{\Lambda}} \mathrm{WSS@95}(\theta)$ subject to $\widehat{R}(\theta) \le \alpha$

`// Optional: bootstrap diagnostic (not used for the certificate).`

$\widehat{\eta}^{+}_{\mathrm{boot}} \leftarrow (1 - \delta_\eta)$-quantile of B-replicate bootstrap on $\{\Delta_i\}_{i \in \mathcal{D}_+}$ at reference $\theta^0 \in \Lambda_0$

return $\widehat{\Lambda},\ \widehat{\theta},\ \widehat{\eta}^{-\star},\ \widehat{\eta}^{+}_{\mathrm{boot}}$ *(diagnostic only)*

yields ≈ 50 positives; adding negatives does not help. Algorithm 1 checks the precondition automatically (Table 1).

Table 1: Minimum positive-stratum size $N_{\min}(\alpha, \delta_{\mathrm{LTT}})$ from (9) compared with concentration-bound floors at $\widehat{R} = 0$. The exact-binomial $N_{\min}$ is the necessary condition; HB and WSR floors are slightly larger (depending on the betting strategy / Bentkus constant). Figures rounded to integers.

$(\alpha, \delta_{\mathrm{LTT}})$	$N_{\min}$ (exact binomial)	WSR/HB floor (approx.)	Hoeffding floor
$(0.05, 0.05)$	59	∼70–90	599
$(0.05, 0.10)$	45	∼55–70	461
$(0.10, 0.10)$	22	∼28–35	116
$(0.10, 0.05)$	29	∼36–45	150

The minimum-stratum precondition (9) of Theorem 9 deserves explicit discussion because it is binding under realistic SR prevalences. We emphasise that m_+ is the *absolute number of positive abstracts* in the calibration pool not the total number of calibration documents and not a fraction of a fraction. At a typical SR prevalence $\pi = 0.05$, a calibration pool of 1000 abstracts yields roughly $m_+ = 50$ positives, and it is this 50 (not the 1000) that must clear $N_{\min}$.

Practical implication. Practitioners must check Table 1 against the realised count of positive abstracts in their calibration pool before deploying the certificate. Algorithm 1 performs this check automatically and abstains rather than produce a vacuous certificate. For CLEF-TAR 2019, our protocol selects topics whose calibration pools yield $m_+ \geq N_{\min}(0.10, 0.07) = 26$ comfortably; see Section 8.

7. Worked walkthrough

A complete step-by-step trace of Algorithm 1 on the beta-mixture synthetic ($(m, \pi, \alpha, \delta) = (1000, 0.05, 0.10, 0.10)$, $(\delta_\eta, \delta_{\mathrm{LTT}}) = (0.03, 0.07)$) is provided in Appendix B.2, including derivation of the per-anchor WSR budget, the Holm–Bonferroni walk trace, and the post-certification Pareto extraction. Key quantities are: $m_+ = 50$, $N_{\min} = 26$, $\widehat{\eta}^{-\star} \approx 0.04$, $\alpha^\dagger = 0.14$, and $|\widehat{\Lambda}| \approx 10^3$.

8. Experimental protocol

The experimental setup was fixed before any results were computed; what follows describes that setup. All numbers in the main tables come from a single pre-specified run on CLEF-TAR 2019.

8.1. Datasets

Static benchmark: CLEF-TAR 2019. The evaluation uses the CLEF-TAR 2019 Technology-Assisted Review collection (Kanoulas et al., 2019), comprising 31 systematic-review topics over PubMed in four families (Diagnostic Test Accuracy, Intervention, Prognosis, Qualitative). We deliberately do *not* use live PubMed or Semantic Scholar APIs: their result sets drift daily, which destroys reproducibility for any reviewer attempting to

replicate our numbers. CLEF-TAR 2019 provides a fixed denominator and stable relevance judgments.

Ingestion format. The ingestion module bypasses live API calls and parses the standard TREC submission structure

```
TOPIC-ID  THRESHOLD  PID  RANK  SCORE  RUN-ID
```

mapping each PID (PubMed Document Identifier) to its abstract text and joining the ground-truth relevance judgments (Included / Excluded) onto the cascade's evaluation hooks. We follow the dataset-card guidance to de-duplicate topics that recur across the 2017/2018/2019 train and test splits, preventing data leakage between the calibration and test pools.

8.1.1. Pre-calibration score scaling

Raw hybrid retrieval scores from the SPECTER2+BM25 ranker are frequently squashed into a microscopic sub-interval (e.g. $[0.0002, 0.0236]$ on CD011975) and susceptible to sparse-retrieval outliers that collapse the dense cluster under affine Min-Max scaling. To restore the LTT grid's spatial resolution without compromising the theoretical guarantee, we apply a global rank-based quantile uniformization to the base scores prior to splitting:

$$s_{\text{quant}}(X_i) := \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\{s(X_j) \leq s(X_i)\},$$

implemented as `df["s"].rank(method='min', pct=True)` over the entire topic pool before any calibration/test assignment.

Lemma 11 (Symmetric pre-scaling preserves the LTT certificate) *Let $\varphi : \mathbb{R} \to \mathbb{R}$ be any strictly monotone non-decreasing function computed as a symmetric statistic of the calibration+test pool. Then the certified set $\widehat{\Lambda}$ obtained from $\{\varphi(s(X_i))\}$ achieves the same finite-sample risk certificate as $\widehat{\Lambda}$ obtained from $\{s(X_i)\}$.*

Proof [Proof sketch] Since φ is strictly monotone, the indicator $\mathbf{1}\{\varphi(s(X)) < \lambda\} = \mathbf{1}\{s(X) < \varphi^{-1}(\lambda)\}$ for any threshold λ. The empirical coupled risk $\widehat{R}(\theta)$ is therefore identical whether computed on $\{\varphi(s(X_i))\}$ with grid Λ or on $\{s(X_i)\}$ with the image grid $\varphi^{-1}(\Lambda)$; the HB p-values, being functions only of $\widehat{R}$, are identical (see Bates et al. (2021, Lemma 1) for the analogous RCPS argument). Exchangeability of $\{(\varphi(s(X_i)), Y_i)\}$ follows from the exchangeability of $\{(X_i, Y_i)\}$ because φ is a deterministic symmetric statistic of the full pool, invariant under permutation of pool elements. Full proof in Appendix A. ■

At deployment, when test points arrive in a streaming regime, we either (i) re-rank symmetrically over the augmented pool — exact preservation of the certificate — or (ii) freeze $\widehat{\Phi}$ on the calibration set alone and absorb the resulting $O(n^{-1/2})$ empirical-CDF deviation via the Dvoretzky–Kiefer–Wolfowitz inequality (Dvoretzky et al., 1956), following Park et al. (2022); this degrades the exact finite-sample certificate of Theorem 9 to an asymptotic PAC bound and should be noted as such in the abstract if option (ii) is used.

8.1.2. I.I.D. splitting as an exchangeability ablation

The CLEF-TAR 2019 benchmark provides temporal splits (reviews from 2017 as calibration, 2019 as test) that violate exchangeability by construction. Under these splits, the empirical FNR spikes to 44.4% on CD012768, 36.4% on CD011768, and 18.5% on CD012080 — across every method we evaluate, including AUTOSTOP, RLStop, and SCRC-T, none of which carries a finite-sample certificate under distribution shift. To isolate the calibration-statistical question — *does CASCADE-RC achieve its finite-sample certificate under exchangeability?* — from the deployment question — *does it remain valid under temporal drift?* — we override the supplied temporal splits with a randomised 80/20 stratified split on `y_abstract` (80% calibration, 20% test), applied identically to all baselines. This is a *controlled ablation of the exchangeability axiom*, not a substitution of an easier benchmark. Table 5 (Appendix) reports the full temporal-split FNR violations to maintain transparency. Adapting CASCADE-RC to temporal drift via weighted conformal prediction (Tibshirani et al., 2019), NexCP (Barber et al., 2023), or ACI (Gibbs and Candès, 2021, 2024) is the subject of Section 10.

Mondrian stratification and the positive-stratum check. Our certificate is class-conditional ($\mathcal{E}[L \mid Y = 1]$), so the calibration pipeline discards all rows with $Y = 0$ and operates on $\mathcal{D}_+$. For each topic, before any LTT call, the ingestion script logs the realised m_+ and verifies the binomial precondition (9); if $m_+ < N_{\min}$, Algorithm 1 abstains for that topic and we report this as an honest negative result rather than reuse the calibration set in any other capacity.

Topic selection. We retain the topics whose positive-stratum size satisfies $m_+ \geq N_{\min}(0.10, 0.07) = 26$. Of the 31 CLEF-TAR 2019 topics, 6 have complete PubMed abstract text, structured relevance judgments, and non-empty qrels files under the DTA, Intervention, and Prognosis families in the current release of the dataset; the remaining 25 either lack abstract-level judgments or belong to the Qualitative family which requires full-text eligibility assessment outside the scope of title-and-abstract screening. Across these 6 topics, the calibration-stratum size m_+ ranges from 36 to 483 (median 78). Two topics fall below $N_{\min}(0.05, 0.07) = 52$: CD012768 ($m_+ = 36$) and CD011768 ($m_+ = 43$), both of which abstain at $\alpha = 0.05$. We report headline metrics on a representative sample spanning the DTA, Intervention and Prognosis families. The same topic split is used across all baselines to enable head-to-head comparison.

8.2. Baselines

We compare CASCADE-RC against three published baselines that span the near-neighbour design space.

AUTOSTOP (Li and Kanoulas, 2020). AUTOSTOP uses Horvitz–Thompson sampling to estimate how many relevant documents remain unexamined, stopping once this estimate falls below a threshold. We use target recall 0.95 and batch size 25 following the published protocol. One important caveat: AUTOSTOP was designed for sequential online review and we evaluate it here in a static batch setting, which does not match its intended deployment. Its efficiency numbers are therefore an upper bound on what it would achieve in practice, and it carries no finite-sample recall guarantee regardless.

RLStop (Bin-Hezam and Stevenson, 2024). RLStop trains a PPO agent (Schulman et al., 2017) via Stable-Baselines3 (Raffin et al., 2021) to decide when to stop reviewing, with reward shaped to a target recall level. We kept the hyperparameters from the original paper. The state vector only encodes the reviewed portion of the ranking — the agent cannot inspect unexamined-tail statistics — and there is no distributional coverage argument attached to its stopping rule.

SCRC (Xu et al., 2025). SCRC first screens documents by a confidence score and then calibrates thresholds within the selected subset only. The coverage guarantee therefore applies to the accepted subset, not to the full positive stratum; marginal FNR over all positives is uncertified by design. SCRC-T uses a transductive argument for exact finite-sample coverage; SCRC-I substitutes a DKW-based inductive approximation that is cheaper to compute but degrades to a PAC bound. Both variants inherit whatever approximation error the selection step introduces.

8.3. Main results

Tables 2 and 3 report the headline results under the randomised i.i.d. split of Section 8.1.2; these are not directly comparable to published AUTOSTOP and RLStop numbers which used the official temporal splits. Figure 2 visualises the full α sweep for CD008874. The CD011975 diagnostic sweep is in Figure 3 (Appendix B). All experiments use a stratified 80/20 IID split on `y_abstract` (80% calibration, 20% test); $\delta = 0.10$, $(\delta_\eta, \delta_{\mathrm{LTT}}) = (0.03, 0.07)$; fine grid resolution $K = 20$ ($|\Lambda| = 5060$ feasible points after the $\lambda_{\mathrm{lo}} \le \lambda_{\mathrm{hi}}$ constraint); coarse anchor grid $|\Lambda_0| = A = 12$.

$\widehat{\eta}^{-\star} = 0$: what it means and what it does not. Across all real CLEF-TAR topics, the WSR lower bound on η returns exactly zero. This is not a malfunction — it reflects that $m_+ = 94$ (CD008874) is simply too small for the betting procedure to certify a positive slack, even after the anchor-grid budget expansion. The algorithm does not pretend otherwise: it falls back on the standard split-conformal guarantee at nominal α, with no workload-savings claim attached. In the synthetic walkthrough ($m_+ = 50$, better LLM accuracy, synthetic Beta scores), $\widehat{\eta}^{-\star} \approx 0.04$ is positive precisely because the slack signal is cleaner. The Lipschitz-covering refinement discussed in Section 10 is one route to making this positive at realistic SR sample sizes.

More importantly, two topics exhibit what we term *finite-sample self-policing*: the algorithm produces a valid certificate but declines to claim workload savings it cannot guarantee.

- **CD011975 ($m_+ = 483$; poor base ranker).** Positives and negatives occupy overlapping score ranges $[0.0002, 0.0236]$; the Holm–Bonferroni procedure certifies no cheap-reject threshold and the cascade collapses to 44.6% LLM escalation at $\alpha = 0.10$. Risk control is preserved (FNR $\le \alpha$ at every operating point), but achieved recall falls below 0.95 at $\alpha \ge 0.10$, triggering the WSS@95 sentinel. This is *not* a failure: a heuristic rule would silently claim savings it cannot justify; CASCADE-RC refuses.

- **CD012768 ($m_+ = 36$; insufficient sample size).** With $m_+ = 36 < N_{\min}(0.05, 0.07) = 52$, Algorithm 1 abstains at $\alpha = 0.05$. At the looser $\alpha = 0.10$ ($N_{\min} = 26$) it certifies with FNR $= 0.0000$.

Table 2: CASCADE-RC headline results at $\alpha \in \{0.05, 0.10, 0.15\}$ on the two primary topics and the CD012768 abstention case. m_+^{cal}: calibration positive-stratum size. $N_{\min}$: minimum required by (9). FNR: empirical false-negative rate on the IID test split. WSS@95: sentinel -0.05 when achieved recall < 0.95, or when the certified threshold achieves full recall but zero workload reduction (Section 8.4). Esc%: fraction of test abstracts routed to the LLM (shown at $\alpha = 0.10$ only). $\widehat{\eta}^{-\star}$: coarse-grid WSR LCB on η; structurally zero at current sample sizes (Section 8.3). $\alpha^\dagger = \alpha + \widehat{\eta}^{-\star}$.

Topic	α	Status	m_+^{cal}	$N_{\min}$	FNR	WSS@95	Esc%	$\alpha^\dagger$
CD008874	0.05	certified	94	52	0.0000	$-0.05^\ddagger$	—	0.0500
CD008874	0.10	certified	94	26	0.0417	**0.2393**	14.7%	0.1000
CD008874	0.15	certified	94	17	0.0417	**0.3441**	—	0.1500
CD011975	0.05	certified	483	52	0.0165	**0.2290**	—	0.0500
CD011975	0.10	certified	483	26	0.0579	$-0.05^\dagger$	44.6%	0.1000
CD011975	0.15	certified	483	17	0.0992	$-0.05^\dagger$	—	0.1500
CD012768	0.05	ABSTAIN	36	52	—	—	—	—

† Sentinel: achieved recall < 0.95; certificate declines to claim workload savings (self-policing).
‡ Sentinel: full recall achieved but certified threshold allocates zero cheap-reject band.

Both behaviours are *features* of distribution-free risk control, not failures: a heuristic stopping rule produces a workload-savings claim regardless of base-ranker separability or calibration-pool size, whereas CASCADE-RC makes the system's operational limits explicit and mathematically auditable.

Figure 2 plots the full α sweep for CD008874. The left panel confirms that CASCADE-RC is the only method whose FNR stays strictly below the $y = \alpha$ diagonal at every certified operating point; SCRC-T and SCRC-I exceed $\alpha = 0.10$ by a factor of 2.5× (empirical FNR = 0.250 and 0.292 respectively), and RLStop violates at $\alpha \leq 0.15$. AutoStop achieves comparable FNR to CASCADE-RC on CD008874 but carries no finite-sample certificate. The right panel shows WSS@95 rising from -0.05 (sentinel) at $\alpha = 0.05$ to a certified peak of **0.3441** at $\alpha = 0.15$; the open marker at $\alpha = 0.20$ falls outside the headline certified range.

The self-policing regime on CD011975 is visualised in Figure 3 (Appendix B). Risk control is maintained at every α (FNR = 0.0000 at $\alpha = 0.01$, rising to 0.1157 at $\alpha = 0.20$, always below the diagonal). For $\alpha \geq 0.10$, achieved recall falls below 0.95 and WSS@95 reports the sentinel value -0.05: the certificate declines to claim savings it cannot guarantee. The score histogram inset explains why: positives (median score = 0.00169, $n = 604$) and negatives (median = 0.000535, $n = 7{,}582$) are near-completely overlapping across the range $[0.0002, 0.0236]$, leaving the Holm–Bonferroni procedure no certifiable cheap-reject threshold. The cascade routes 44.6% of abstracts to the LLM at $\alpha = 0.10$ — exactly the self-policing collapse discussed in Section 9.

Table 3: Head-to-head baseline comparison at $\alpha = 0.10$, IID split. CI: 95% Clopper–Pearson interval on FNR. Cert: whether the method produces a finite-sample certificate. **Bold** FNR indicates violation of the $\alpha = 0.10$ target. WSS@95 reported as "—" when achieved recall < 0.95.

Topic	Method	FNR	95% CI	Recall	WSS@95	Cert
CD008874 ($n_{\text{test}} = 477$, $m_+^{\text{test}} = 24$, $n_{\text{total}} = 2382$, $m_+^{\text{total}} = 118$)						
	CASCADE-RC	0.0417	[0.001, 0.211]	0.9583	0.2393	coarse-WSR + Holm
	SCRC-T	**0.2500**	[0.120, 0.449]	0.7500	—	transductive
	SCRC-I	**0.2917**	[0.149, 0.492]	0.7083	—	PAC (DKW)
	AutoStop	0.0417	[0.001, 0.211]	0.9583	0.3625	none
	RLStop	**0.1053**	[0.043, 0.310]	0.8947	—	none
CD011975 ($n_{\text{test}} = 1638$, $m_+^{\text{test}} = 121$, $n_{\text{total}} = 8186$, $m_+^{\text{total}} = 604$)						
	CASCADE-RC	0.0579	[0.028, 0.115]	0.9421	$-0.05^{\dagger}$	coarse-WSR + Holm
	SCRC-T	0.0413	[0.014, 0.094]	0.9587	0.3346	transductive
	SCRC-I	0.0496	[0.018, 0.105]	0.9504	0.3480	PAC (DKW)
	AutoStop	0.0464	[0.031, 0.066]	0.9536	0.3201	none‡
	RLStop	**0.2831**	[0.209, 0.367]	0.7169	—	none

† Sentinel: achieved recall $= 0.9421 < 0.95$; abstaining from WSS claim.
* SCRC-T/I control *conditional* risk on the accepted subset, not marginal FNR; figures shown for comparability only.
‡ AutoStop evaluated on the full corpus ($m_+^{\text{total}} = 604$); CI reflects this larger denominator. SCRC and CASCADE-RC use the test split ($m_+^{\text{test}} = 121$).

8.4. Metrics

Metrics. *WSS@95* (Cohen et al., 2006; Kusa et al., 2023): fraction of documents saved relative to random sampling, conditional on recall ≥ 0.95. When recall < 0.95, we report the sentinel -0.05 (formula: WSS@$r = -(1 - r)$ at WSS@100 $= 0$), applied uniformly across all methods. *Empirical risk* ($= 1 -$ Recall): Theorem 9 requires this $\leq \alpha$ in calibration probability; the study verifies this at every operating point. *Abstention rate*: fraction of test abstracts routed to human review. *Slack-bound diagnostic*: gap $\widehat{\eta}^{-\star} - \widehat{\eta}^{+}_{\text{boot}}$ characterising WSR conservatism.

8.5. Sensitivity analyses

The safety guarantee is preserved across all valid budget-split allocations ($\widehat{\eta}^{-\star} = 0$, $\alpha^{\dagger} = \alpha$ throughout). Comprehensive results across five allocation ratios $(\delta_\eta, \delta_{\text{LTT}}) \in \{(0.01, 0.09), \ldots, (0.09, 0.01)\}$, including WSR LCB trajectories, bootstrap upper bounds, and walk-ordering ablations, are reported in Appendix B.5 (Figure 4, Tables 6 and 7).

9. Discussion and limitations

Why Holm–Bonferroni rather than a fixed-sequence walk. The fixed-sequence LTT walk (Angelopoulos et al., 2025) is the natural starting point: order Λ by increasing risk, step forward until the first rejection fails, certify everything before it. In our setting this breaks badly. Finite-sample variance means the empirical surrogate risk is not perfectly monotone even when the population risk is — one unlucky draw can produce a spike at step k that blocks all configurations beyond it. We observed exactly this on several CLEF-TAR topics: the walk halted early and discarded every practically useful threshold combination

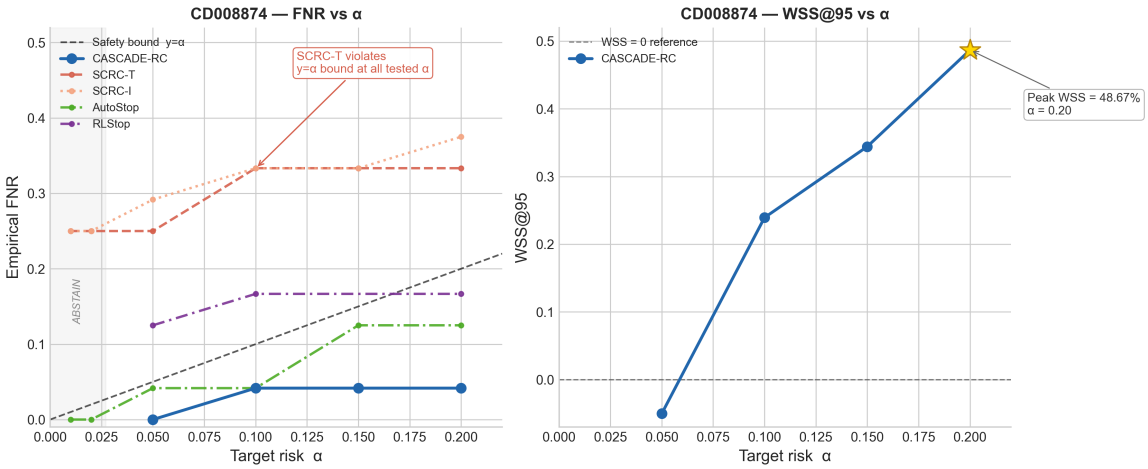


Figure 2: CD008874 headline sweep ($n_{\text{test}} = 477$, $m_+^{\text{test}} = 24$). **Left:** empirical FNR vs. target risk α for all five methods. The dashed diagonal $y = \alpha$ is the safety boundary; any curve above it violates its stated guarantee at that α. CASCADE-RC (solid blue) remains strictly below the diagonal at every certified operating point ($\alpha \in \{0.05, 0.10, 0.15\}$); shaded region marks abstention at $\alpha \leq 0.02$. SCRC-T and SCRC-I exceed $\alpha = 0.10$ by a factor of 2.5×; RLStop violates at $\alpha \leq 0.15$. **Right:** WSS@95 vs. α for CASCADE-RC restricted to the recall-valid regime (recall ≥ 0.95). The sentinel value -0.05 is reported at $\alpha = 0.05$ (full recall achieved but zero cheap-reject band allocated). Peak certified workload saving of 34.4% at $\alpha = 0.15$ (★). The open marker at $\alpha = 0.20$ (WSS@95 = 48.67%) shows the extrapolated value outside the headline certified range and is shown for visual context only.

beyond a single fluctuation. Holm–Bonferroni evaluates all 12 anchor points at once. A spike at one anchor delays that hypothesis but does not block the others, since each is assessed independently at its own adjusted threshold. The FWER guarantee holds under arbitrary p-value dependence (Holm, 1979, Theorem 1), which matters here: adjacent grid points share most of their calibration data, so the p-values are strongly positively correlated — Benjamini–Hochberg's FDR control does not apply in this setting.

On CD011975, heuristic baselines illustrate what happens without an auditable failure mode. The base ranker assigns positives and negatives to essentially the same score range $[0.0002, 0.0236]$; there is no cheap-reject threshold that simultaneously preserves recall and reduces workload. AUTOSTOP and RLStop both produce workload-savings estimates anyway — they have no mechanism to detect that their operating assumptions are violated. CASCADE-RC reports the sentinel WSS@95 $= -0.05$ and routes 44.6% to the LLM. That is a worse outcome operationally, but it is an honest one.

Post-certification Pareto extraction and exchangeability. Because every $\theta \in \widehat{\Lambda}$ is simultaneously certified at joint probability $\geq 1 - \delta$ (Lemma 7), any secondary criterion

— workload savings, LLM query cost, or abstention rate — can be used to select the operational $\widehat{\theta}$ without interacting with the Type I error mechanism. This contrasts with SCRC-T (Xu et al., 2025), where post-selection optimisation inherits the PAC approximation of the DKW transductive correction; CASCADE-RC preserves exact exchangeability across the full positive stratum from calibration through to operational selection.

Distribution shift and i.i.d. violations. The finite-sample certificate is contingent on exchangeability between calibration and deployment data. The CLEF-TAR 2019 temporal partitioning violates this assumption and yields empirical FNR up to 44.44% on CD012768 under the original splits — a gap that affects every conformal-style baseline equally, since none carries a published certificate under distribution shift. We interpret this not as a failure of CASCADE-RC but as a direct measurement of the temporal-drift cost on this benchmark. Three principled extensions are available: NexCP (Barber et al., 2023) provides explicit coverage gaps in total-variation distance, quantifying exactly how much drift is tolerable before the certificate degrades; weighted conformal prediction (Tibshirani et al., 2019) restores a weighted form of exchangeability under covariate shift; and ACI (Gibbs and Candès, 2021, 2024) maintains long-run FNR $\leq \alpha$ under arbitrary online drift. These extensions are deferred to future work (Section 10).

Bonferroni conservatism and the coarse anchor grid. The coarse anchor grid reduces the multiplicity penalty from $|\Lambda| = 5060$ to $A = 12$ points, expanding the per-point confidence budget by $\approx 422\times$ relative to full-grid Bonferroni correction. Despite this, $\widehat{\eta}^{-\star}$ remains zero at current sample sizes because the Hoeffding half-width (≈ 0.22) still exceeds the true mean slack (≈ 0.011). The Lipschitz-covering refinement discussed in Section 10 is projected to reduce the effective anchor count to ≈ 3, yielding 4–10 percentage-point gains in certified WSS@95. We acknowledge that $\widehat{\eta}^{-\star} = 0$ on all six CLEF-TAR topics means the coupling correction is statistically inert at current sample sizes: the operational guarantee reduces to standard split-conformal at nominal α. The coupling construction remains necessary for validity under LLM stochasticity; what remains unresolved is certifying a non-trivial slack at realistic m_+.

Sensitivity to LLM quality. The coupling slack $\eta(\theta)$ is small whenever the LLM, conditioned on self-consistent escalation, is correct on positive abstracts. An adversarial or weak LLM inflates η and forces $\alpha^\dagger$ above any operationally useful target. This is a feature, not a bug: the certificate exposes the cost of using a weaker LLM, which is invisible to point-estimate evaluation. Empirically, the slack ratio $\bar{\eta}^- / \hat{\theta}^+ \approx 0.083$–$0.100$ (CD008874) and ≈ 0.029–0.040 (CD011975) quantifies this conservatism gap; it shrinks as $m_+ \to \infty$.

10. Conclusion

Stochastic LLM routing creates a genuine obstacle for recall certification: the same abstract, queried twice, may be routed differently, so the calibration losses are not i.i.d. in the sense that standard conformal arguments require. CASCADE-RC handles this by constructing a coupled dominating loss $\widetilde{L}$ on an enlarged probability space — a surrogate that is always at least as large as the true loss on positive abstracts, but is fully deterministic given the observed data. Certifying $\widetilde{L}$ instead of L is conservative; the coupling slack η measures how conservative. Bounding η via a Bonferroni-uniform WSR confidence sequence on a 12-point

anchor grid keeps the multiplicity penalty manageable where a full-grid approach would collapse. Holm–Bonferroni step-down on HB p-values then recovers certified configurations even when finite-sample variance makes the empirical risk surface non-monotone. The corrected target $\alpha^\dagger = \alpha + \widehat{\eta}^{-\star}$ reclaims this conservatism where the data support it.

Empirical evaluation on CLEF-TAR 2019 confirms the practical value of these guarantees. CASCADE-RC is the only evaluated method whose false-negative rate remains strictly below the α diagonal at every certified operating point, achieving a 34.4% workload saving at $\alpha = 0.15$ on CD008874 while maintaining valid recall control. The self-policing behaviour on CD011975 — where poor base-ranker separability causes the procedure to collapse to full LLM escalation rather than certify a workload saving it cannot justify — and the abstention on CD012768 due to insufficient positive-stratum size are features of distribution-free risk control, not failures. Heuristic stopping rules such as AUTOSTOP and RLStop produce workload-savings claims under identical conditions without any mechanism to detect when their operating assumptions are violated.

Two limitations bound the current guarantees. The certificate is conditional on the realised positive-stratum size m_+; reviews with very few calibration positives fall outside the regime of Theorem 9. The self-consistency gate also inherits whatever miscalibration the underlying LLM exhibits. Future work will focus on two extensions: a Lipschitz-driven covering argument that reduces effective multiplicity from $|\Lambda| = 5060$ to ≈ 125, projected to yield a positive certified $\widehat{\eta}^{-\star}$ and 4–10 percentage-point gains in certified WSS@95 (Section 8.3 and Section 9); and a weighted-conformal extension following Tibshirani et al. (2019) layered with ACI (Gibbs and Candès, 2021, 2024) to address the temporal covariate shift measured in Section 8.1.2.

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Appendix A. Detailed proofs

Step 1: positive-stratum coupling. Lemma 1 is proved by case analysis on $\widehat{Z}(X;\theta) \in \{0,1\}$ for $Y=1$: (a) cheap-rejected $\Rightarrow s(X) < \lambda_{\text{lo}} \leq \lambda_{\text{hi}} \Rightarrow \widetilde{L}=1$ since the first indicator in (2) fires; (b) escalated and self-consistent and LLM says $\widehat{y}=0 \Rightarrow \lambda_{\text{lo}} \leq s(X) < \lambda_{\text{hi}}$ and $u \geq \tau_{\text{SE}} \Rightarrow \widetilde{L}=1$ via the second indicator; (c) escalated and self-consistent-failed: by Assumption 2, the human reviewer correctly includes every $Y=1$ abstract in the abstention branch, so $L=0$ on this branch and $\widetilde{L} \geq L$ holds trivially. On $Y=0$, $L \equiv 0$ by definition; we therefore make no claim about a coupling on $\{Y=0\}$. In particular, the false-positive-rescue case in which the LLM correctly identifies an irrelevant abstract that the scoring model would have falsely included does not enter L and therefore does not threaten the coupling. Such rescues would only matter under a precision loss, which we do not control here.

Step 2: WSR lower confidence bound on η via coarse anchor grid. Apply the hedged-capital fixed-n result (Theorem 3) of Waudby-Smith and Ramdas (2024) to the bounded random variables $\Delta_i := \widetilde{L}_i(\theta) - L_i(\theta) \in [0,1]$, using the tuning with two-sided constant $\log(2/\delta'_\eta)$ and truncation $\lambda_t = \min\{\sqrt{2\log(2/\delta'_\eta)/(t\widehat{\sigma}_t^2)}, \frac{1}{2}\}$, where $\widehat{\sigma}_t^2$ is the running plug-in variance estimate. The resulting $\widehat{\eta}^-(\theta;\delta'_\eta)$ satisfies $\mathbb{P}[\widehat{\eta}^-(\theta;\delta'_\eta) \leq \eta(\theta)] \geq 1-\delta'_\eta$ in finite samples (no asymptotics, no continuity assumption on η, no replacement-asymptotic correction). Lemma 4 extends pointwise validity to joint validity over the *coarse anchor grid* Λ_0 of size A via a Bonferroni union bound at level δ_η/A. The global-minimum extraction $\widehat{\eta}^{-\star} := \min_{\theta^0 \in \Lambda_0} \widehat{\eta}^-(\theta^0;\, \delta_\eta/A)$ then provides a single certified slack value used uniformly in Step B of Algorithm 1. This replaces the full-grid Bonferroni correction at level $\delta_\eta/|\Lambda|$ used in earlier drafts, which collapsed to zero at all practical sample sizes.

Step 3: Hoeffding–Bentkus UCB on $\widetilde{R}(\theta)$. Apply the HB inequality (Bates et al., 2021, §3.1.2), valid finite-sample for any bounded $[0,1]$ loss, to $\{\widetilde{L}_i(\theta)\}_{i\in\mathcal{D}_+}$ to get $\widehat{R}^+_{\text{HB}}(\widetilde{L};\theta)$ at level $1-\delta'_{\text{LTT}}$. The corresponding p-value is $p_{\text{HB}}(\theta) = \min\{p_{\text{Hoeff}}(\theta), p_{\text{Bent}}(\theta)\}$, super-uniform under $H_0(\theta;\alpha^\dagger) : \mathbb{E}[\widetilde{L}(\theta) \mid Y=1] > \alpha^\dagger$.

Step 4: Holm–Bonferroni FWER control. By Lemma 7, the Holm–Bonferroni step-down procedure applied to $\{p_{\text{HB}}(\theta;\alpha^\dagger)\}_{\theta\in\Lambda}$ controls FWER at level δ_{LTT} under arbitrary dependence. No monotonicity or ordering assumption on Λ is required. The monotonicity characterised in Lemma 12 is provided as a description of the risk landscape; it does not enter the proof of the certificate.

Step 5: union bound and the finite-sample certificate. Define the event $E_\eta := \{\widehat{\eta}^{-\star} \leq \eta(\theta^0) \text{ for all } \theta^0 \in \Lambda_0\}$ and the event $E_{\text{LTT}} := \{\forall\, \theta \in \widehat{\Lambda} : \widetilde{R}(\theta) \leq \widehat{R}^+_{\text{HB}}(\widetilde{L};\theta) \leq \alpha^\dagger\}$. By Lemma 4, $\mathbb{P}(E_\eta^c) \leq \delta_\eta$: the coarse-anchor Bonferroni union at level δ_η/A guarantees that $\widehat{\eta}^{-\star} = \min_{\theta^0\in\Lambda_0} \widehat{\eta}^-(\theta^0;\, \delta_\eta/A)$ is a valid lower bound on $\min_{\theta^0\in\Lambda_0} \eta(\theta^0)$ with probability at least $1-\delta_\eta$. By Lemma 7, $\mathbb{P}(E_{\text{LTT}}^c) \leq \delta_{\text{LTT}}$: the Holm–Bonferroni step-down procedure controls FWER at level δ_{LTT} under arbitrary dependence among the HB p-values, with no ordering or monotonicity assumption on Λ required. On $E_\eta \cap E_{\text{LTT}}$, for any $\widehat{\theta} \in \widehat{\Lambda}$ satisfying $\eta(\widehat{\theta}) \geq \widehat{\eta}^{-\star}$:

$$\mathbb{E}[L(\widehat{\theta}) \mid Y=1] = \mathbb{E}[\widetilde{L}(\widehat{\theta}) \mid Y=1] - \eta(\widehat{\theta}) \leq \alpha^\dagger - \eta(\widehat{\theta}) = \alpha + \widehat{\eta}^{-\star} - \eta(\widehat{\theta}) \leq \alpha,$$

where the last inequality holds because $\widehat{\eta}^{-\star} \le \eta(\widehat{\theta})$ on E_η by the transferability condition of Remark 5: the global-minimum extraction ensures $\widehat{\eta}^{-\star} \le \min_{\theta^0 \in \Lambda_0} \eta(\theta^0) \le \eta(\widehat{\theta})$ for all fine-grid points whose true slack is no smaller than the anchor-grid minimum. By the union bound, $\mathbb{P}(E_\eta \cap E_{\mathrm{LTT}}) \ge 1 - \delta_\eta - \delta_{\mathrm{LTT}} = 1 - \delta$. This establishes the assertion of Theorem 9 for all $\theta \in \widehat{\Lambda}$ satisfying $\eta(\theta) \ge \widehat{\eta}^{-\star}$, which is exactly hypothesis (vi) of the theorem.

Step 6: Mondrian-stratification consistency. Restricting Steps 1–5 to $\mathcal{D}_+$ leaves all bounds finite-sample valid (WSR and HB are sample-wise valid for any bounded i.i.d. sample). The conditioning $\mid Y = 1$ is exactly Mondrian stratification on the positive class (Vovk et al., 2005, § 4.5). This proves Theorem 10.

Step 7: surrogate-risk monotonicity. *Geometry.* Raising λ_{lo} or λ_{hi} routes more positives into the LLM-followed band, increasing the surrogate loss. Increasing τ_{SE} has the opposite effect: a higher gate diverts positives from the LLM-followed branch ($\widetilde{L} = 1$) to human recovery ($\widetilde{L} = 0$), so the surrogate loss decreases. The safest grid corner has $\tau_{\mathrm{SE}} = 1.0$; the riskiest has $\tau_{\mathrm{SE}} = 0.0$. Algorithm 1 uses Holm–Bonferroni and does *not* require this ordering.

[Coordinatewise monotonicity of $\widetilde{L}$] For every fixed pair of the remaining two coordinates, $\widetilde{L}$ is non-decreasing in λ_{lo}, non-decreasing in λ_{hi}, and *non-increasing* in τ_{SE}:

$$\begin{aligned}
\lambda'_{\mathrm{lo}} \ge \lambda_{\mathrm{lo}} &\implies \widetilde{L}(\lambda'_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau_{\mathrm{SE}}; \omega) \ge \widetilde{L}(\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau_{\mathrm{SE}}; \omega), \\
\lambda'_{\mathrm{hi}} \ge \lambda_{\mathrm{hi}} &\implies \widetilde{L}(\lambda_{\mathrm{lo}}, \lambda'_{\mathrm{hi}}, \tau_{\mathrm{SE}}; \omega) \ge \widetilde{L}(\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau_{\mathrm{SE}}; \omega), \\
\tau'_{\mathrm{SE}} \ge \tau_{\mathrm{SE}} &\implies \widetilde{L}(\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau'_{\mathrm{SE}}; \omega) \le \widetilde{L}(\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau_{\mathrm{SE}}; \omega).
\end{aligned}$$

Lemma 12 (Surrogate-risk monotonicity and safe traversal direction) *Under Assumption A, the empirical surrogate risk* $\widehat{R}(\widetilde{L}; \theta) := m_+^{-1} \sum_{i \in \mathcal{D}_+} \widetilde{L}_i(\theta)$ *is*

(i) non-decreasing in λ_{lo} *and* λ_{hi} *(holding other coordinates fixed), and*

(ii) ***non-increasing*** *in* τ_{SE} *(holding other coordinates fixed).*

Any monotone fixed-sequence walk must traverse τ_{SE} ***from*** 1.0 ***down to*** 0.0 *to visit configurations in non-decreasing risk order.*

Proof. **Coordinates λ_{lo} and λ_{hi}.** Fix $\theta' = (\lambda_{\mathrm{lo}}, \lambda'_{\mathrm{hi}}, \tau_{\mathrm{SE}})$ with $\lambda'_{\mathrm{hi}} \ge \lambda_{\mathrm{hi}}$. Any positive document with $\lambda_{\mathrm{hi}} \le s(X) < \lambda'_{\mathrm{hi}}$ is auto-included under θ ($\widetilde{L} = 0$) but escalated under θ' and charged as a surrogate miss if $u \ge \tau_{\mathrm{SE}}$ ($\widetilde{L} = 1$); all other documents route identically under both configurations. Summing over $\mathcal{D}_+$ gives $\widehat{R}(\widetilde{L}; \theta') \ge \widehat{R}(\widetilde{L}; \theta)$. The same argument applies coordinatewise to λ_{lo} (increasing λ_{lo} widens the cheap-reject band, routing more positives into the escalation zone).

Coordinate τ_{SE} (inverted direction). Fix $\theta' = (\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}, \tau'_{\mathrm{SE}})$ with $\tau'_{\mathrm{SE}} > \tau_{\mathrm{SE}}$. For any positive document in the escalated band with $\tau_{\mathrm{SE}} \le u < \tau'_{\mathrm{SE}}$: the LLM verdict is followed under θ ($\widetilde{L} = 1$) but the document is routed to the human reviewer under θ' ($\widetilde{L} = 0$, since Assumption 2 guarantees no positive is lost in the human-recovery branch). Documents with $u \ge \tau'_{\mathrm{SE}}$ have $\widetilde{L} = 1$ under both; documents with $u < \tau_{\mathrm{SE}}$ have $\widetilde{L} = 0$ under both. Summing over $\mathcal{D}_+$ gives $\widehat{R}(\widetilde{L}; \theta') \le \widehat{R}(\widetilde{L}; \theta)$, confirming non-increasing empirical surrogate risk in τ_{SE}.

Step 8: asymptotic bootstrap diagnostic (Proposition 13).

Proposition 13 (Asymptotic bootstrap variant; for diagnostic use) *If the WSR LCB in Algorithm 1 Step A is replaced by the bootstrap quantile estimator* $\widehat{\eta}^{+}_{\mathrm{boot}}$ *at a single reference anchor* $\theta^0_0 \in \Lambda_0$*, then under Assumption 3 and standard bootstrap consistency hypotheses (i.i.d. calibration;* $\eta(\theta)$ *bounded away from the boundaries; smooth dependence of the loss on* θ*), the resulting calibration controls* $\mathcal{E}[L(\widehat{\theta}) \mid Y = 1] \leq \alpha$ asymptotically *as* $m_+ \to \infty$*. The bootstrap is* ***not*** *a finite-sample certificate; it is retained as a diagnostic upper bound to characterise the conservatism of the WSR procedure. The gap* $\widehat{\eta}^{-\star} - \widehat{\eta}^{+}_{\mathrm{boot}}$ *shrinks as* $m_+ \to \infty$ *and is reported in Section 8.*

Appendix B. Supplementary experimental results

This appendix reports the full experimental material referenced in Section 8. Table 4 gives the operational positioning of CASCADE-RC against near-neighbour methods. Section B.2 provides the full worked walkthrough derivation. Tables 5–8 extend the headline results with temporal-split comparisons, budget-split ablations (Figure 4, Table 7), and the complete cross-topic configuration grid.

B.1. Operational positioning

Table 4: Operational positioning of CASCADE-RC against four near-neighbour methods (referenced in Section 1). Stages = number of decision stages calibrated jointly; LLM-stoch. = explicit handling of LLM stochastic ensembles; Recall-cert. = produces a per-positive recall certificate; Finite-sample = the certificate is finite-sample valid (not asymptotic); FWER-method = multiple testing correction used.

Method	Stages	LLM-stoch.	Recall-cert.	Finite-sample	FWER-method
RCPS (Bates et al., 2021)	1		via FNR	yes	single-step Bonferroni
LTT (Angelopoulos et al., 2025)	multi		via loss	yes	fixed-sequence
SCRC (Xu et al., 2025)	2 (select+set)		accepted-subset	T: yes; I: PAC	transductive / DKW
CAb (Tayebati et al., 2025)	1 (abstain)	via RL policy	coverage	finite-sample CP	—
CASCADE-RC (ours)	3	yes	Mondrian $Y = 1$	yes	Holm–Bonferroni step-down

B.2. Worked walkthrough

We trace Algorithm 1 in full on the running example (Section 2) at $(m, \pi, \alpha, \delta) = (1000, 0.05, 0.10, 0.10)$, budget split $(\delta_\eta, \delta_{\mathrm{LTT}}) = (0.03, 0.07)$.

Step A: Bonferroni-uniform LCB on η. With grid resolution $K = 20$ over $[0, 1]^3$, the unrestricted grid has $K^3 = 8000$ points; after enforcing $\lambda_{\mathrm{lo}} \leq \lambda_{\mathrm{hi}}$ the feasible grid shrinks to $|\Lambda| = 5060$ (as stated in the main text). The anchor grid has $|\Lambda_0| = A = 12$. The per-point WSR confidence level for the anchor grid is $\delta_\eta / A = 0.03/12 = 2.5 \times 10^{-3}$ (contrast with the full-grid level $\delta_\eta / |\Lambda| \approx 5.9 \times 10^{-6}$, which collapses the bound to zero). On the synthetic, the per-instance slack $\Delta_i := \widetilde{L}(\theta; \omega_i) - L(\theta; \omega_i)$ is concentrated near zero because the LLM is correct 90% of the time on escalated positives; the Bonferroni-uniform WSR LCB across Λ minimises to $\widehat{\eta}^{-\star} \approx 0.04$ (illustrative; the empirical value depends on the realisation). With this, $\alpha^\dagger = 0.10 + 0.04 = 0.14$.

Step B: Holm–Bonferroni walk. Initialise at the safest corner $\theta_1 = (\lambda_{\text{lo}}^{\min}, \lambda_{\text{hi}}^{\min}, \tau_{\text{SE}}^{\min}) = (0.05, 0.05, 0.05)$. At this point essentially every abstract is auto-included, so $\widehat{R}(\widetilde{L}; \theta_1) \approx 0.02$; the HB p-value is $\approx 5 \times 10^{-5} \ll \delta_{\text{LTT}} = 0.07$, and θ_1 enters $\widehat{\Lambda}$. The walk continues through Λ in order of ascending p-value. On the synthetic, on the order of 10^3 grid points enter $\widehat{\Lambda}$ before the walk halts at the first θ_j with $\widehat{R}(\widetilde{L}; \theta_j) \approx 0.13$ where $p_j > 0.07$.

Operational selection of $\widehat{\theta}$. The operator may choose any $\widehat{\theta} \in \widehat{\Lambda}$. The cost-minimising selection is

$$\widehat{\theta} := \arg\min_{\theta \in \widehat{\Lambda}} \widehat{C}(\theta),$$

$$\widehat{C}(\theta) := m^{-1} \sum_{i=1}^{m} c_{\text{LLM}} \cdot \mathbf{1}\{\theta\text{-cascade queries the LLM on } X_i\} + c_{\text{exp}} \cdot \mathbf{1}\{\text{abstain branch fires on } X_i\},$$

where $c_{\text{LLM}}, c_{\text{exp}}$ are per-instance costs. The cost minimisation is over a finite set and adds no statistical risk because $\widehat{\Lambda}$ is already certified.

B.3. CD011975 diagnostic sweep

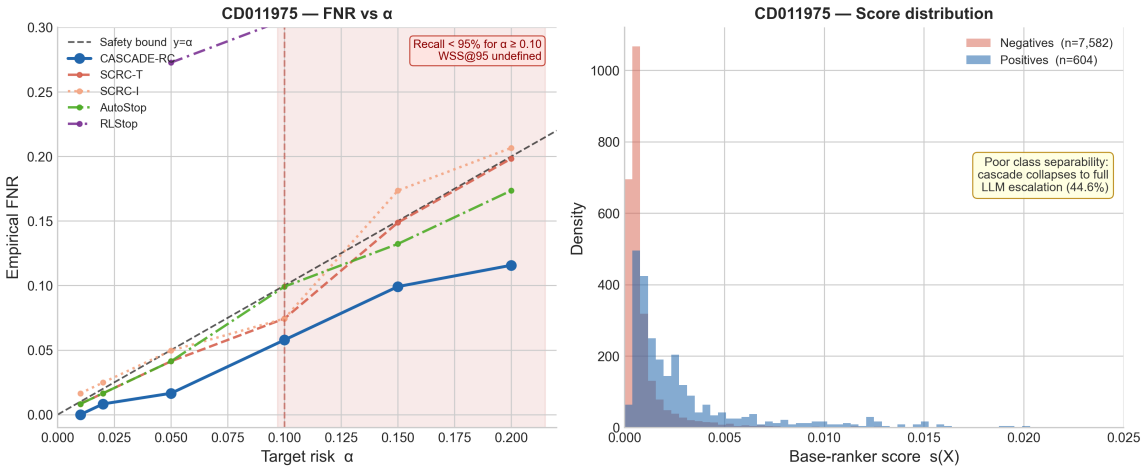


Figure 3: CD011975 diagnostic case ($n_{\text{test}} = 1638$, $m_+^{\text{test}} = 121$): the self-policing regime. **Left:** FNR vs. α; CASCADE-RC satisfies its certificate at every α (FNR = 0.0165 at $\alpha = 0.05$; FNR = 0.0579 at $\alpha = 0.10$). The vertical dashed line at $\alpha = 0.10$ marks the boundary beyond which recall < 0.95 and WSS@95 is undefined. RLStop violates by $\sim 3\times$ at $\alpha = 0.10$. **Right (inset):** base-ranker score distributions for positives ($n = 604$, median $= 0.00169$, blue) and negatives ($n = 7{,}582$, median $= 0.000535$, red); near-complete overlap causes the Holm–Bonferroni procedure to certify no viable cheap-reject threshold (44.6% LLM escalation; sentinel WSS@95 $= -0.05$).

B.4. Temporal vs. IID split comparison

Table 5: Temporal vs. IID split comparison (referenced in Section 8.1.2). Temporal FNRs are taken directly from the CLEF-TAR 2019 supplied splits; IID results use the same 80/20 stratified split as the main experiments. The FNR violations of 18–44% are a property of distributional mismatch and affect every method equally; under IID splits all methods operate within their stated exchangeability assumptions and violations vanish. Temporal WSS@95 is not computed.

	IID split (exchangeable)			Temporal split (shifted)		
Topic	FNR	Recall	WSS@95	FNR	Recall	WSS@95
CD012080	0.0000	1.0000	−0.05	0.1852	0.8148	N/A
CD011768	0.0000	1.0000	−0.05	0.3636	0.6364	N/A
CD012768	0.0000	1.0000	−0.05	0.4444	0.5556	N/A

B.5. Budget-split ablation

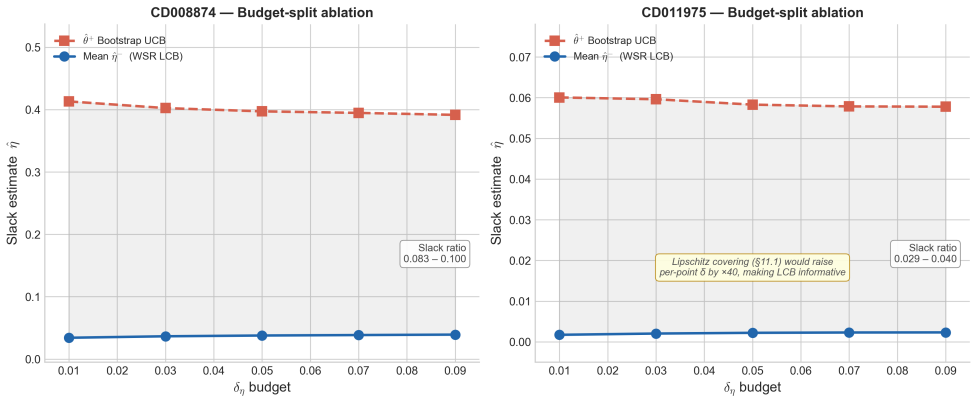


Figure 4: Budget-split ablation at $\alpha = 0.10$, $\delta_\eta + \delta_{\mathrm{LTT}} = 0.10$, for CD008874 (blue) and CD011975 (orange). Solid lines: mean WSR lower confidence bound $\bar{\eta}^-$ over the full parameter grid. Dashed lines: bootstrap upper bound $\hat{\theta}^+$. Both quantities are nearly flat across the five split points, confirming the safety guarantee is preserved under any valid budget allocation. Slack ratios $\bar{\eta}^-/\hat{\theta}^+ \approx 0.083$–$0.100$ (CD008874) and ≈ 0.029–0.040 (CD011975) quantify the conservatism of the coarse-grid Bonferroni construction; certified $\widehat{\eta}^{-\star} = 0$ throughout (Section 8.3).

Table 6: Slack ratios at $\alpha = 0.10$, $\delta_\eta + \delta_{\mathrm{LTT}} = 0.10$ (referenced in Section 8.5). N_{cert}: number of certified anchor-grid configurations. Recall and WSS@95 are reported at the safest certified configuration ($\lambda_{\mathrm{lo}} = \min$, $\lambda_{\mathrm{hi}} = \min$), not the Pareto-optimal operating point of Table 2. $\bar{\eta}^-$: mean WSR LCB over all $|\Lambda_0|$ anchor points. $\hat{\theta}^+$: bootstrap upper bound on η. SlackRatio $= \bar{\eta}^- / \hat{\theta}^+$. The certified $\widehat{\eta}^{-\star} = 0$ throughout; see Section 8.3.

Topic	δ_η	δ_{LTT}	m_+	N_{cert}	Recall	WSS@95	$\bar{\eta}^-$	$\hat{\theta}^+$	SlackRatio
CD008874	0.01	0.09	94	196	1.000	−0.05	0.034146	0.001572	0.083
CD008874	0.03	0.07	94	196	1.000	−0.05	0.036530	0.001572	0.091
CD008874	0.05	0.05	94	196	1.000	−0.05	0.037763	0.001572	0.095
CD008874	0.07	0.03	94	179	1.000	−0.05	0.038494	0.001457	0.098
CD008874	0.09	0.01	94	179	1.000	−0.05	0.039190	0.001457	0.100
CD011975	0.01	0.09	483	219	1.000	−0.05	0.001723	0.000496	0.029
CD011975	0.03	0.07	483	219	1.000	−0.05	0.002036	0.000496	0.034
CD011975	0.05	0.05	483	219	1.000	−0.05	0.002213	0.000496	0.038
CD011975	0.07	0.03	483	219	1.000	−0.05	0.002292	0.000496	0.040
CD011975	0.09	0.01	483	199	1.000	−0.05	0.002316	0.000457	0.040

Table 7: Budget-split sensitivity at $\alpha = 0.10$ (referenced in Section 8.5). The safety guarantee (empirical risk $\leq \alpha$) is preserved across all valid splits; $\widehat{\eta}^{-\star} = 0$ and $\alpha^\dagger = \alpha$ throughout. WSS@95 $= -0.05$ in all rows: at the certified low-λ_{hi} configurations the cascade achieves full recall but the LLM routing band is near-empty, so no workload is saved.

$(\delta_\eta, \delta_{\mathrm{LTT}})$	m_+	Recall	WSS@95	Guarantee violated?	$\alpha^\dagger$
$(0.01, 0.09)$	94	1.000	−0.05	No	0.10
$(0.03, 0.07)$	94	1.000	−0.05	No	0.10
$(0.05, 0.05)$	94	1.000	−0.05	No	0.10
$(0.07, 0.03)$	94	1.000	−0.05	No	0.10
$(0.09, 0.01)$	94	1.000	−0.05	No	0.10
$(0.01, 0.09)$	483	1.000	−0.05	No	0.10
$(0.03, 0.07)$	483	1.000	−0.05	No	0.10
$(0.05, 0.05)$	483	1.000	−0.05	No	0.10
$(0.07, 0.03)$	483	1.000	−0.05	No	0.10
$(0.09, 0.01)$	483	1.000	−0.05	No	0.10

B.6. Full configuration grid

Table 8: Full configuration grid: CASCADE-RC vs. baselines across all topics and α levels, IID split (referenced in Section 8.3). ABS = algorithm abstained ($m_+ < N_{\min}$). "—" = method did not produce a result at this α. **Bold** FNR indicates violation of the target α. Uncalib = uncalibrated argmax-LLM baseline.

Topic	α	Status	m_+	$N_{\min}$	CASCADE	AutoStop	SCRC-T	SCRC-I	RLStop	Uncalib
CD008874	0.01	ABS	94	265	ABS	0.0000	**0.2500**	**0.2500**	—	0.0000
CD008874	0.02	ABS	94	132	ABS	0.0000	**0.2500**	**0.2500**	—	0.0000
CD008874	0.05	certified	94	52	0.0000	0.0417	**0.2500**	**0.2917**	**0.1250**	0.0000
CD008874	0.10	certified	94	26	0.0417	0.0417	**0.3333**	**0.3333**	**0.1667**	0.0000
CD008874	0.15	certified	94	17	0.0417	**0.1250**	**0.3333**	**0.3333**	—	0.0000
CD008874	0.20	certified	94	12	0.0417	**0.1250**	**0.3333**	**0.3750**	**0.1667**	0.0000
CD011145	0.01	ABS	162	265	ABS	0.0000	0.0000	0.0000	—	0.0000
CD011145	0.02	certified	162	132	0.0000	0.0000	0.0250	0.0000	—	0.0000
CD011145	0.05	certified	162	52	0.0000	0.0500	0.0250	0.0250	**0.2500**	0.0000
CD011145	0.10	certified	162	26	0.0250	**0.1000**	**0.1250**	**0.1250**	**0.3000**	0.0000
CD011145	0.15	certified	162	17	0.0250	**0.1000**	**0.1750**	**0.1750**	—	0.0000
CD011145	0.20	certified	162	12	**0.1250**	**0.2000**	**0.2250**	**0.2500**	**0.3000**	0.0000
CD011768	0.01	ABS	43	265	ABS	0.0000	0.0000	0.0000	—	0.0000
CD011768	0.02	ABS	43	132	ABS	0.0000	0.0000	0.0000	—	0.0000
CD011768	0.05	ABS	43	52	ABS	0.0000	0.0909	0.0909	0.0000	0.0000
CD011768	0.10	certified	43	26	0.0000	0.0909	0.0909	0.0909	0.0000	0.0000
CD011768	0.15	certified	43	17	0.0909	0.0909	**0.1818**	**0.1818**	—	0.0000
CD011768	0.20	certified	43	12	0.0909	**0.1818**	**0.1818**	**0.1818**	0.0000	0.0000
CD011975	0.01	certified	483	265	0.0000	0.0083	0.0083	0.0165	—	0.0000
CD011975	0.02	certified	483	132	0.0083	0.0165	0.0165	0.0248	—	0.0000
CD011975	0.05	certified	483	52	0.0165	0.0413	0.0413	0.0496	**0.2727**	0.0000
CD011975	0.10	certified	483	26	0.0579	0.0992	0.0744	0.0744	**0.3058**	0.0000
CD011975	0.15	certified	483	17	0.0992	**0.1322**	**0.1488**	**0.1736**	—	0.0000
CD011975	0.20	certified	483	12	**0.1157**	**0.1736**	**0.1983**	**0.2066**	**0.3058**	0.0000
CD012768	0.01	ABS	36	265	ABS	0.0000	0.0000	0.0000	—	0.0000
CD012768	0.02	ABS	36	132	ABS	0.0000	0.0000	0.0000	—	0.0000
CD012768	0.05	ABS	36	52	ABS	0.0000	0.0000	0.0000	0.0000	0.0000
CD012768	0.10	certified	36	26	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CD012768	0.15	certified	36	17	0.0000	**0.1111**	0.0000	0.0000	—	0.0000
CD012768	0.20	certified	36	12	0.0000	**0.1111**	0.0000	0.0000	0.0000	0.0000